

8.6

a.)

$$H_0: \sigma_m^2 = \sigma_f^2$$

$$H_1: \sigma_m^2 \neq \sigma_f^2$$

$$df_m = 577 - 4 = 573$$

$$df_f = 1000 - 577 - 4 = 419$$

$$\alpha = 5\%$$

$$\hat{\sigma}_m^2 = \frac{SSE}{df} = \frac{97161,9174}{573} = 169,57$$

$$\hat{\sigma}_f^2 = (12,024)^2 = 144,58$$

$$F = \frac{\hat{\sigma}_m^2}{\hat{\sigma}_f^2} = \frac{169,57}{144,58} = 1,17$$

$$F_{0,975}(573, 419) \approx 1,197$$

Reject region: $F > 1,197$

$\Rightarrow$ Do not reject $H_0: \sigma_m^2 = \sigma_f^2$ at 5% level.

$\Rightarrow$ There is no evidence at 5% level to conclude that the error variance differs between males and females.

b)

$$H_0: \sigma_{\text{Marriage}}^2 = \sigma_{\text{single}}^2$$

$$H_1: \sigma_{\text{Marriage}}^2 > \sigma_{\text{single}}^2$$

$$df_{\text{marriage}} = 600 - 5 = 595$$

$$df_{\text{single}} = 100 - 5 = 95$$

$$\alpha = 5\%$$

$$\hat{\sigma}_{\text{marriage}}^2 = \frac{100.703,0471}{595} = 169,25$$

$$\hat{\sigma}_{\text{single}}^2 = \frac{56.231,0382}{95} = 142,36$$

$$F = \frac{\hat{\sigma}_{\text{marriage}}^2}{\hat{\sigma}_{\text{single}}^2} = \frac{169,25}{142,36} = 1,19$$

$$F_{0,95}(595, 95) = 1,165$$

Reject region: $F > 1,165$

(one side test)

$\Rightarrow$ Reject $H_0: \sigma_{\text{marriage}}^2 = \sigma_{\text{single}}^2$ at 5% level

$\Rightarrow$ The married individuals have larger wage error variance than the single one at 5% level.

c)

$$NR^2 = 59,03$$

$$\text{Test statistic } \chi^2 = NR^2 \sim \chi_{14}^2$$

$$\alpha = 5\% \Rightarrow \chi_{0,95,14}^2 = 9,49$$

$$NR^2 > \chi_{0,95,14}^2 \Rightarrow \text{Reject homoskedasticity at 5\% level.}$$

$\Rightarrow$ There is heteroskedasticity at 5% level.

$\Rightarrow$ This doesn't prove the issue in part (b) that the error variation is different for married and unmarried one. Because this test only shows the error variance related to at least one of the variables (EDUC, EXPER, METRO, FEMALE), not other factors outside the model.

d)

4 originals: EDUC, EXPER, METRO, FEMALE

2 Squares: EDUC², EXPER²

6 cross-products: 402 = 6

$$\Rightarrow df = 4 + 2 + 6 = 12$$

$$\alpha = 5\% \Rightarrow \chi_{0,95,12}^2 = 21,03$$

$$NR^2 = 78,82$$

$$\Rightarrow NR^2 > \chi_{0,95,12}^2$$

$\Rightarrow$ Reject homoskedasticity at 5%

$\Rightarrow$ There is heteroskedasticity at 5%

e)

- Wider : intercept, EDUC
- Narrower : EXPER, METRO, FEMALE

⇒ There is no inconsistency. The RSE could be either larger or smaller under the context of heteroskedasticity.

f)

- No, it is compatible
 - Part (b) concerns whether the error variance differs between married and single individuals.
 - Part (f) t-test on married examines whether marital status affect the conditional avg. wage
- ⇒ 2 different aspects.
- It is possible that married with $RSE = 1.0$ still consistent with the finding in (b).

8.16

a.

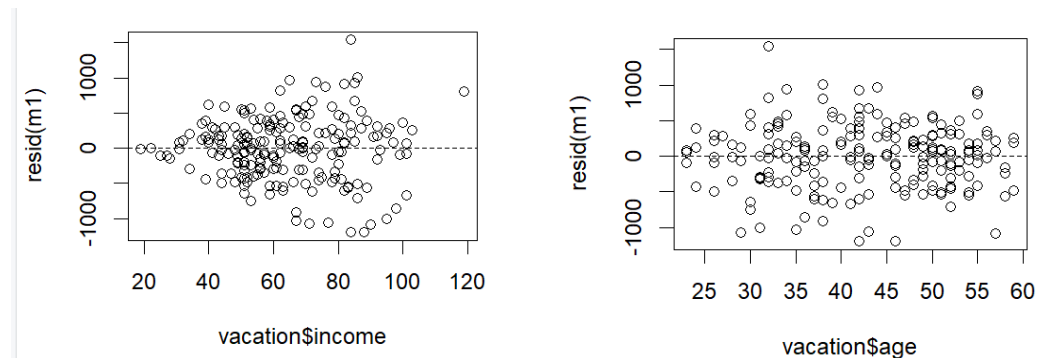
```
> confint(m1,"kids", level = 0.95)
      2.5 %      97.5 %
kids -135.3298 -28.32302
```

In the repeated sampling, about 95% intervals constructed this way will contain the true value of parameter KIDS.

b.

The residual plot against INCOME shows the spread increasing with income, which suggests heteroskedasticity. The residual plot against AGE does not show a strong variance pattern.

c.



$$H_0: \sigma^2_{\text{high}} = \sigma^2_{\text{low}}$$

$$H_1: \sigma^2_{\text{high}} > \sigma^2_{\text{low}}$$

Goldfeld–Quandt test:

```
> sse_low <- sum(resid(m_low)^2)
> sse_low
[1] 8750040
> sse_high <- sum(resid(m_high)^2)
> sse_high
[1] 27160654
> F <- (sse_high/df_high)/ (sse_low/df_low)
> F
[1] 3.104061
> f_crit <- qf(0.95, df_high, df_low)
> f_crit
[1] 1.428617
```

The test statistic is higher than the critical value, we reject the null Hypothesis H_0 at 5% level.

There is evidence to conclude that there is heteroskedasticity and the error variance increases with income.

d.

```
—  
> ci_kids_rb  
      kids      kids  
-139.32297 -24.32986
```

95% interval estimate is [-139.32297 ; -24. 32986]. The CI is wider than the CI in part a, which we expect there is heteroskedasticity.

e.

```
> ci_kids_gls_trad  
      2.5 %      97.5 %  
kids -119.8945 -33.71808  
> ci_kids_gls_rb  
      kids      kids  
-121.41339 -32.19919  
OLS CI: [-135.33 ; -28.33]
```

Robust OLS CI: [-139.31 ; -24.35]

GLS CI: about [-119.9 ; -33.7]

So the GLS interval is narrower than both OLS intervals, meaning it is more precise under the maintained variance assumption $\sigma_i^2 = \sigma^2 INCOME_i^2$